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Review article

A survey on non-terrestrial quantum networking: Challenges and trends

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ABSTRACT

Satellites and aerial platforms can be significant in the development of new telecommunications networks, allowing to deploy networks with extremely high performance even in remote areas. Recently, considering the development of quantum communication technologies, many studies are concentrated on Free Space Optic (FSO) Non-Terrestrial links, also considering that the studies performed so far on Optical Fibers (OFs) have not pointed out significant results, especially in terms of coverage. Moreover, the decrease of launch costs and the use of new satellite technologies such as CubeSats make the deployment of satellite constellations affordable. Thanks to the deployment of Non-Terrestrial Quantum Networks (NTQNs), the novel services offered by the Quantum Internet (QI) could be made easily accessible to all. In particular, Quantum Computers (QCs) can perform beyond the capabilities of any classical processor, and the interconnection of these devices on a global scale makes it possible to achieve extremely high computational capabilities, thus opening the way to new applications, such as Quantum Cloud (QCloud). Therefore, the recent studies in this area motivate this survey, which in addition to providing some base concept of quantum information, describes the current technologies needed to create quantum NT backbones. Specifically, this survey examines some studies over the period 2010–2023, which are related to the realization of quantum communications using satellites and other aerial platforms such as High Altitude Platforms (HAPS) and quantum drones. Some of the architectures that are described in this survey make use of the Software-Defined Networking (SDN) paradigm, which can be envisaged as a key enabler for the management of heterogeneous QNs in general. Furthermore, through the description of various experiments, the paper shows the applications that can be enabled by the future quantum NT backbones. In particular, in addition to improving the security of current systems through the Quantum Key Distribution (QKD), it could be possible to interconnect clusters of QCs over long distances and create extremely reliable positioning systems. Finally, the paper outlines some possible future developments.

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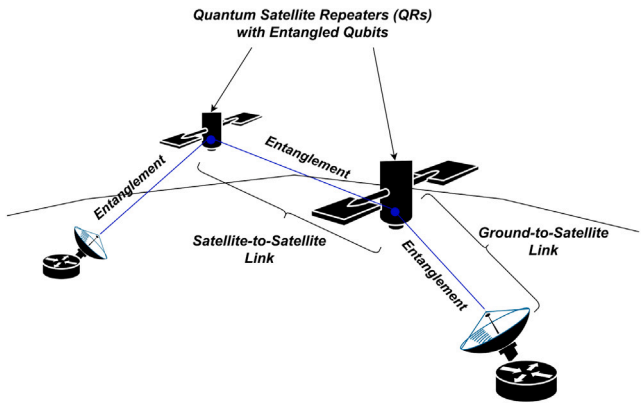

Fig. 1. Links required to perform a quantum satellite communication. Entanglement is generated on each single link.

1. Introduction

The exponential increase of devices supply and the demand for advanced services require the development of novel telecommunications networks [1]. In particular, Non-Terrestrial Networks (NTNs), which include satellite networks and aerial platforms, provide an extremely suitable solution, also considering that they allow connecting areas underserved and where it is difficult to deploy a terrestrial infrastructure, thus complementing, e.g., terrestrial fifth-generation (5G) and sixth generation (6G) networks [2,3].

Recently, quantum technologies are emerging as a fundamental paradigm for creating new generation networks beyond the 6G [4], which are based on the quantum entanglement [5] and quantum teleportation [6] phenomena, with particular consideration to the use of satellite links [7–10]. In a quantum communication two different channels are required, i.e., a classical channel and a quantum channel [11,12].

As a matter of fact, obtaining the entanglement between two remote devices is still very difficult considering Optical Fiber (OF) links [13]. Therefore, devices called Quantum Repeaters (QRs) are used in order to extend the range of entanglement between two devices through a specific protocol that requires several steps [14,15]. Specifically, the QRs divide the entire communication channel into multiple segments, making it possible to create pairs of entangled particles between adjacent nodes by transmitting photons entangled with their Quantum Memories (QMs) [9,16]. In fact, if the transmitted modes are entangled and light is absorbed by the set of atoms of which the memory is composed, the quantum correlations can be mapped on the collective superposition of the two final states of the memories [17,18].

Furthermore, considering the difficulties encountered in the conservation of Bell states due to the interaction of particles with matter that induces *decoherence*, i.e., the loss of coupling in time, it is necessary to resort to other communication media than OFs ones [34]. A promising solution consists of using satellite technologies in order to use the space vacuum as communication medium as much as possible since it has better characteristics [35,36] (see Fig. 1).

A communication through a Quantum Satellite Backbone (QSB) can be performed considering the use of two different links [37,38], i.e., Satellite-to-Satellite links and Forward-and-Return (F&R) satellite links. In particular, the F&R links are very critical because they pass through the lower layers of the atmosphere introducing diffraction losses, absorption, scattering, beam wandering, beam spreading, and scintillation effects. However, in fair weather conditions, the entanglement can be preserved [39] and, therefore, the quantum NTNs technology could potentially allow the creation of a *global quantum communication network* to connect any two points on Earth, reducing propagation losses than OFs. Numerous experiments have already been conducted and consider the use of small-size and low-cost satellites, i.e., CubeSats [8,40], using the Free Space Optic (FSO) technology [41]. The availability of CubeSats and the reduction of launch costs that occurred with the beginning of the SpaceX initiative have produced growth of satellite constellation programs and could be determinant also in the development of future Quantum Satellite Networks (QSNs) [41]. These satellites are usually placed in Low Earth Orbit (LEO), allowing them to reduce their inter-distance and the distance to Ground Stations (GSs), which constitute a significant point in quantum communications. Nevertheless, the most significant drawback is the high speed of the satellites rotating around Earth with speed between 5 to 10 km/s [42,43], which makes difficult achieving a very accurate pointing during signal transmission. Furthermore, the communication can only be performed during specific time windows, due to the limited flyover time of the satellites [30].

Instead, networks of High Altitude Platforms (HAPS) or other aerial platforms such as drones can constitute a key element in the realization of these networks also because they can be deployed more rapidly, significantly reducing the inter-distance and costs. Moreover, as reported in [44] the level of miniaturization that has now been achieved makes it possible to equip these carriers with very high-performance quantum devices. However, given their limited coverage these platforms can only be used on a regional scale and integration with satellite networks is therefore necessary. Thus, it is important to find a compromise between ease of management and coverage.

These Quantum Networks (QNs) could support the realization of the so-called Quantum Internet (QI), that compared to the classic Internet, enables many other applications several of which related to security [45]. However, the achievable level of security provided by the physical laws on which QNs are based [46,47], is merely theoretical and specific defense methods involving combination with classical protocols to avert possible attacks have already been developed [48]. Furthermore, additional benefits that can be obtained from such a network consist in some applications, including distributed computing that can enable advanced Cloud platforms, allowing the quick processing of huge amounts of data. The final objective is to interconnect clusters of Quantum Computers (QCs), i.e. an ensemble of Quantum Data Centers (QDCs) that work in a coordinated fashion obtaining a further increase of computational capacity, an objective that can be achieved through the use of specific QSNs [49].

Many studies described in this survey also regard the use of the Software-Defined Networking (SDN) paradigm applied to QSNs, with the aim of interconnecting QDCs. In fact, given the problems experienced in creating effective path selection and management protocols, the SDN has been recognized as essential for the development of future QNs. For this reason, it has already been used in the Quantum Key Distribution (QKD) application, thus originating the so-called Quantum SDN (QSDN) [50,51]. In particular, the Control Plane (CP) is primarily responsible for the management of the devices belonging to the Data Plane (DP) that can create Einstein–Podolsky–Rosen (EPR) pairs between them. Then, in the case of QSDN, the DP traffic is a flow of entangled particles sent through a quantum channel, whereas the CP traffic is composed of both the classical information needed for the teleportation process and the background protocol traffic that is related to path selection and entanglement generation management

Table 1
Comparison of this survey with other related surveys.

Survey	Quantum mechanics	Quantum entanglement	Quantum teleportation	QKD	Quantum satellite backbones	Quantum HAPS and drone backbones	Quantum SDN	Quantum distributed computing and cloud	Quantum positioning systems
This overview	✔	✔	✔	✔	✔	✔	✔	✔	✔
Singh Amoldeep et al. (2021) [19]	✓	✓	✓	✓					
Angela Sara Cacciapuoti et al. (2020) [20]	✓	✓	✓	✓					
Stefano Pirandola et al. (2020) [21]		✓		✓					
Feihu Xu et al. (2020) [22]		✓		✓					
Valerio Scarani et al. (2019) [23]		✓	✓	✓					
Laszlo Gyongyosi and Sandor Imre (2019) [24]	✓								
Laszlo Gyongyosi et al. (2019) [25]				✓					
Nedasadat Hosseinidehaj et al. (2019) [26]	✓	✓		✓					
Stephanie Wehner et al. (2018) [27]		✓	✓	✓					
Laszlo Gyongyosi et al. (2018) [28]	✓	✓		✓					
Zunaira Babar et al. (2018) [29]	✓	✓							
Robert Bedington et al. (2017) [30]		✓	✓	✓					
Stefano Pirandola et al. (2015) [31]		✓		✓					
Hoi-Kwong Lo et al. (2014) [32]		✓	✓	✓					
Gerardo Adesso and Fabrizio Illuminati (2007) [33]	✓	✓	✓	✓					

procedures [8]. In some experiments that are illustrated in this survey, the SDN has been used with the aim of managing the satellite network and the operations necessary to establish entanglement between two GSs [7–9]. Furthermore, a network of quantum satellites could be deployed around Earth in order to improve the current positioning systems considerably [52,53]. In fact, the traditional Global Navigation Satellites Systems (GNSSs) present an error in the range of 10 to 15 m, which may be unacceptable for many applications [54–56] as geodesy, gravitational wave observation, and synthetic aperture optical astronomy [57,58]. However, many of these errors are related to clock synchronization, and they can be mitigated by QSNs, allowing to achieve an unprecedented level of accuracy limited only by the fundamental bounds of quantum metrology, which is the study of making high-resolution and highly sensitive measurements of physical parameters using quantum theory [59,60]. Specifically, this result can be achieved by creating a satellite-based atomic clock network that jointly allocates their respective resources in a global network involving entangled quantum states [52,56]. Moreover, considering that the quantum entanglement can be used in order to quadratically improve the sensitivity of measurements [61], a QSN can also be used in order to perform precise measurements related to phenomena involving quantum physics [52,53].

The main objective of this survey is to explore the most recent studies related to quantum NTNs and possible applications, also considering their integration with terrestrial QNs. Specifically, the feasibility of global quantum applications and services requires heterogeneous networking, and in this context, the NTN segment is fundamental. Therefore, we explored the proposed solutions by highlighting that specific network management policies are required to control the complex operations necessary to establish an E2E quantum connection. Furthermore, we investigated the application of the SDN principle considered crucial for the management of quantum NTN segments.

As shown in Table 1, differently from existing papers, this survey includes several studies on quantum NT backbones over the period 2010–2023 and the enabled applications. Specifically, the issues explored in this paper are the following:

- QSNs.
- Quantum HAPS and drone networks.
- NT QRs and backbones coordinated through the application of the SDN paradigm.
- Quantum distributed computing and Cloud architectures.
- Atomic clocks synchronization and Quantum Positioning System (QPS).

To the best of our knowledge, the use of satellite networks and aerial platforms in quantum communications is an open issue, and consequently, this survey provides a remarkable contribution to this field. Moreover, this survey mainly discusses realistic experiments available in the literature.

In Fig. 2 is depicted the structure of the paper. Section 2 explains the motivations for the use of quantum NTNs, it describes the devices and the evolution of NTNs considering also the use of SDN paradigm. Section 3 describes some feasibility studies related to the implementation of QNs using different types of aerial platforms and the introduction of SDN approach to control them. Section 4 discusses through several experiments the applications that can be enabled through a QSN. Finally,

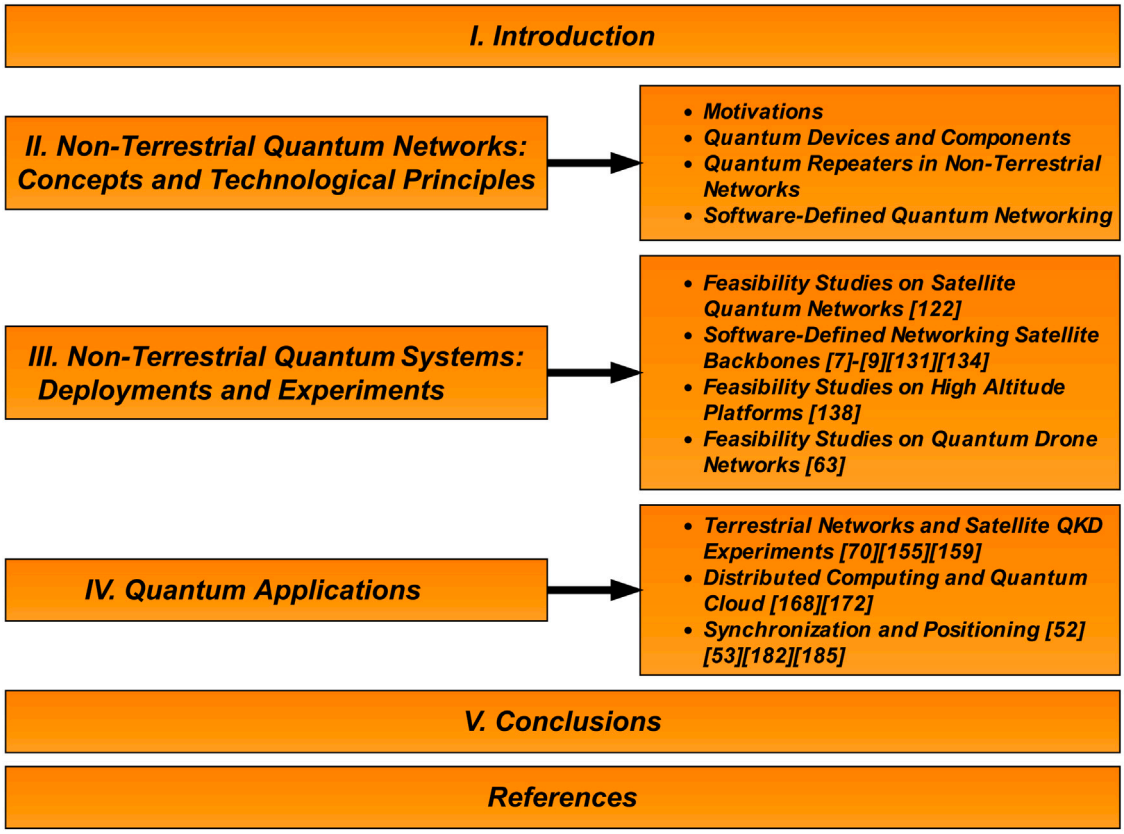

Fig. 2. Sections that compose the paper. The Figure includes the most significant studies cited in this Survey for each topic.

Section 5 concludes the document and identifies the critical issues on which it is necessary to work for the development of integrated Non-Terrestrial Quantum Networks (NTQNs).

2. Non-terrestrial quantum networks: Concepts and technological principles

This Section examines the technological principles for the creation of quantum NTNs. Specifically, besides illustrating the reasons that conduct to the employment of FSO links, the Section describes the basic components and the motivations related to the application of SDN for the design of NT quantum systems.

2.1. Motivations

As already pointed out in [13,62], for the specific case of OFs the probability of success to generate entangled photons between two adjacent nodes decreases exponentially with the distance d and can be characterized using the following model:

$$p_{i,j}(d) = \frac{1}{2}(p_g \eta_h \eta_t)^2 e^{-\frac{d}{L_a}} \tag{1}$$

Specifically, p_g is the probability of success to generate EPR pairs, while η_h and η_t are the detector efficiencies at herald and telecom wavelength. The factor $\frac{1}{2}$ means that only two of the four Bell states can be unambiguously identified performing a Bell State Measurement (BSM) [13,62]. Furthermore, L_a is the attenuation length of a typical OF.

As explained in [13], the average time required to generate a remote entanglement between two adjacent nodes n_i and n_j depends on the probability expressed in (1) and results:

$$T_{i,j} = \frac{p_{i,j}(d)T^s_{i,j} + (1 - p_{i,j}(d))T^f_{i,j}}{p_{i,j}(d)} \tag{2}$$

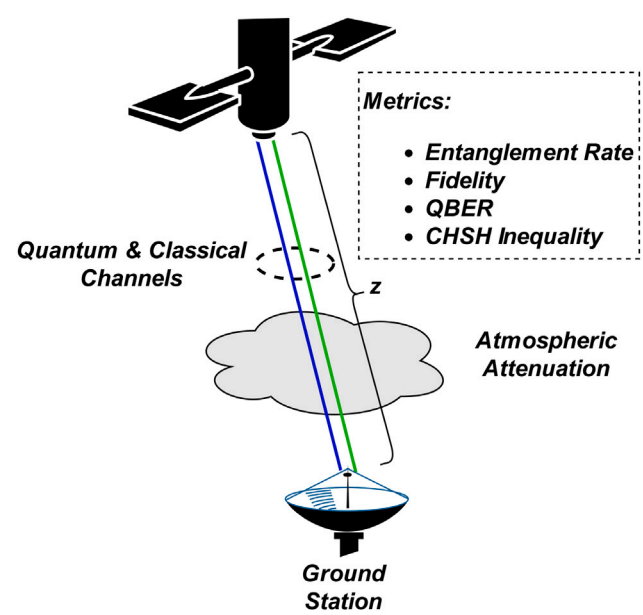

Fig. 3. Representation of the link between a GS and a Satellite in a quantum communication context.

where $T^s_{i,j}$ and $T^f_{i,j}$ are, respectively, the average time required for the successful and unsuccessful attempt. Therefore, the entanglement rate between two adjacent nodes n_i and n_j is equal to:

$$R = \frac{1}{T_{i,j}} \tag{3}$$

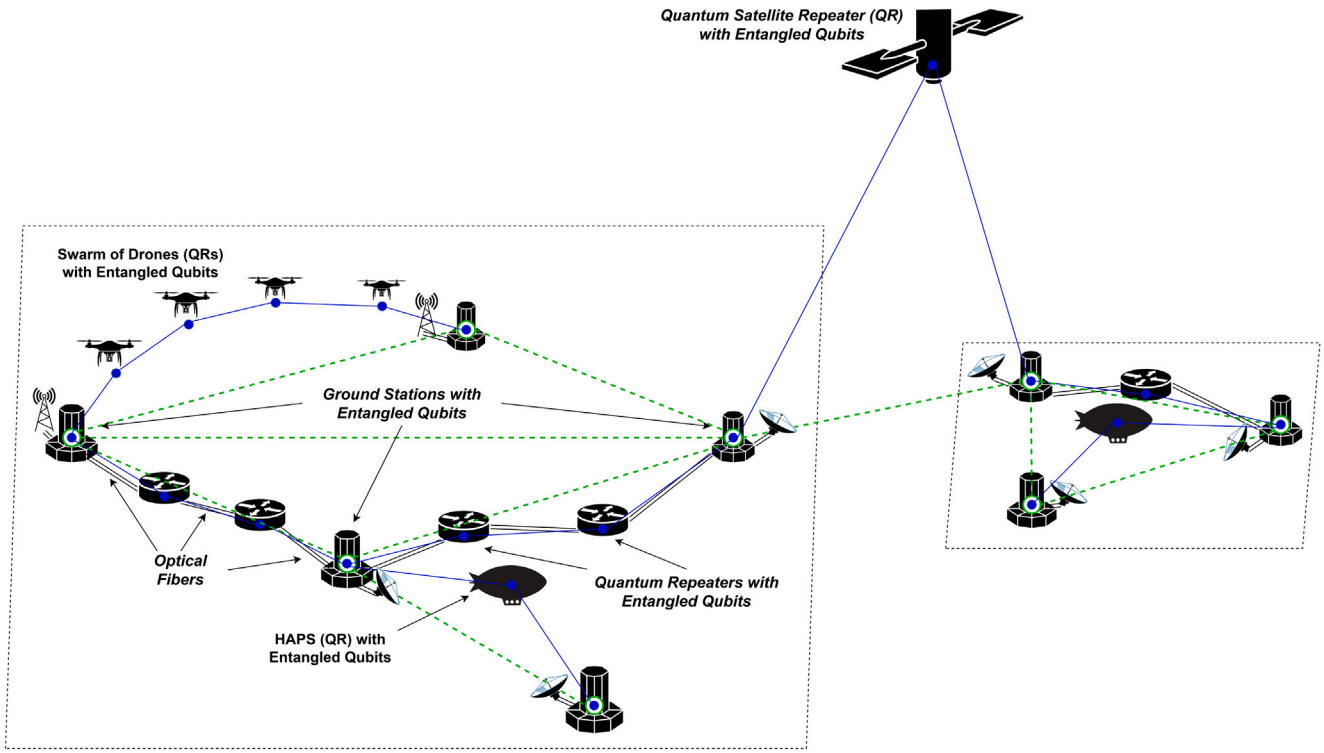

Fig. 4. An example of a mixed scenario in which GSs with entangled qubits are interconnected through QRs installed on different platforms.

As stated in [63], these parameters are strongly influenced by the characteristics and length of the communication channel.

To overcome the distance limit, multiple QRs on the path between the transmitter and receiver have to be deployed. Differently from conventional repeaters used in classical communication, QRs cannot clone quantum signals [64]. These devices are equipped with QMs, allowing to map of a quantum state onto an ensemble of atoms and then recovering it in its original shape [65].

Considering the results in [13], which shows a significant decrease in the rate at the increasing of OF link length, the deployment of a terrestrial QN would require a high number of QRs.

Free-space quantum links have been recently considered, since they allow to obtain performance significantly higher compared to OFs ones [66–68]. For instance, in [67] a quantum communication between two places that were 143 km apart was established, creating a quantum link in the lower atmosphere. In [69], an Italian research group demonstrates the feasibility of QKD protocols along a Ground-to-Satellite channel. Furthermore, as described in [70], with the Micius satellite a quantum communication between locations on Earth at a distance of maximum 7600 km has been demonstrated.

However, free-space communication depends on: (i) free-space attenuation bounded by the squared distance between the GS and the satellite, (ii) the size of the telescopes, (iii) the atmospheric boundary, and (iv) the beam pointing, as shown in [71,72]. In addition, atmospheric turbulence introduces phenomena such as beam wondering, beam-spreading, and scintillation. Nevertheless, as stated in [63] it is possible to achieve fidelity and entanglement rate values higher than OFs considering atmospheric links under specific meteorological conditions, modeled by the Mie scattering expressed in [26], which in turns includes Kruse's model for the scattering coefficient (see Fig. 3). Further, as specified in [13], considering an OF link of 50 km length, it is possible to achieve an entanglement rate equal to 20 Bellpairs/s. Conversely, in free-space communications, under specific visibility conditions, it is possible to achieve the same target value over one order of magnitude longer links, as shown in Table 2, where only the effects of atmospheric propagation for a wavelength of 1550 nm are considered.

Table 2
FSO link length that allows to obtain an entanglement rate of 20 Bellpairs/s considering different atmospheric conditions.

Atmospheric visibility [km]	Attenuation [dB/km]	FSO link length [km]
25	0.177	56.587
50	0.065	154.431
75	0.043	231.647
100	0.032	308.862

Similar values of attenuation under different atmospheric conditions are also reported in [71,73,74]. Moreover, as reported in [30], even using the best OFs technologies, the maximum link lengths to achieve a rate sufficient for a practical use are on the order of 100 km.

The beam-wandering causes a random deviation of the beam-center (x_c, y_c) along the x and y coordinates from the center of the receiver's aperture around a target point (x_d, y_d) following a two dimensional Gaussian distribution:

$$p(x_c, y_c) = \frac{1}{2\pi\sigma_w^2} e^{-\frac{(x_c-x_d)^2+(y_c-y_d)^2}{2\sigma_w^2}} \tag{4}$$

where σ_w is the beam-wandering standard deviation. Other phenomena induced by atmospheric turbulence are (i) beam-spreading, which is a broadening of the beam width and (ii) scintillation, that induces fluctuations in the beam intensity [26].

As described in [75], the path of the satellites is determined by its altitude and orbit. In particular, the parameters which define an orbit are eccentricity and inclination [76]. Therefore, numerous objects performing typical QR operations can be deployed at different orbits and altitudes. Recently, multiple layers in which to deploy communication apparatus that involve terrestrial and satellite segments have been considered, taking into account the technological developments of aerial and miniaturized satellites. Specifically, most of the conducted studies have involved Low Earth Orbit (LEO), HAPS and Low Altitude Platforms (LAP). As regards quantum communications, some of these configurations are still under investigation. To guarantee the best

possible performance on an E2E path, the features of the inter-LEO, inter-HAP, and inter-LAP links have to be studied, also considering hybrid architectures such as those depicted in Fig. 4.

2.2. Quantum devices and components

As explained in [77], in general, a QR must be equipped with two interfaces, an interface for quantum communication to exchange Bell pairs and a classical interface dedicated to the traffic exchanged on the classical channel. In [12], a scheme of a quantum communication system is described. In particular, a quantum device consists of a quantum photon source and a quantum photon detector, both driven by dedicated hardware. The quantum light source and detectors are specific semiconductor devices for the generation of entangled photons, which can be realized with different technologies, as discussed in [78]. Moreover, as reported in [79], a significant building block that composes a QR is the QM. Therefore, a distributed QC network requires specific memory capabilities: mapping of the state of the photons onto memory, the possibility to store and operate on the memory state, and to convert the state of memory back into light for further processing [17]. As explained in [80], the performance of a QM is characterized by the following parameters:

- Fidelity: represents the correlation between the input state and the retrieved state.
- Efficiency: it is given by the ratio between the probability of having acquired a photon and the probability of being able to recover it.
- Storage duration: a QM should store an input quantum state for as long as possible. However, the storage duration is limited by thermal effects, atomic interactions, external electric or magnetic field noises.

These parameters depend on the technology, and, in particular, storage times vary from a few nanoseconds to tens of seconds [81].

Although given the above considerations, the use of NTN platforms appears to be an excellent alternative to terrestrial quantum networks, the integration of a QR on a satellite introduces several technical challenges [82]. QM technology is still under development, and architecture ensuring high efficiency, long coherence times, and multi-mode functionality is still difficult and expensive to realize [80]. Nevertheless, many of the required technologies have been already individually developed, and, some of them, experimented in the space environment. For instance, dilution refrigerators, which are devices that provide cooling at extremely low temperatures, have been tested in microgravity conditions [83]. In fact, the operating temperature of a QM has been achieved in different experiments onboard the International Space Station [84,85], in which were also evaluated the ability to reach ultra-high vacuum, stable operation of lasers and microwave-radio sources, and sizeable artificial magnetic fields. Recently, a good level of miniaturization has been achieved regarding quantum devices. As a matter of fact, in [44] is presented a 19-inch rack Quantum Computing (QCing) demonstrator composed of 40 Ca^{+} optical qubits realized with linear Paul trap technology [86].

In order to generate the entanglement among multiple devices, there are three schemes [20,87] depicted in Fig. 5 and described in the following part:

- At mid-point: a source of entangled photons is positioned between the two involved nodes to which it sends the generated Bell pair.
- At source: in this scheme, the photonic source is included in one of the two nodes, which sends to the other one photons entangled with the qubits of its memory.
- At both end-points: in this approach, both nodes are equipped with photonic sources and send to a specific detector photons entangled with the qubits of their memory. Therefore, the detector measures the two qubits; in this manner, an entangled quantum state between the two remote qubits is created.

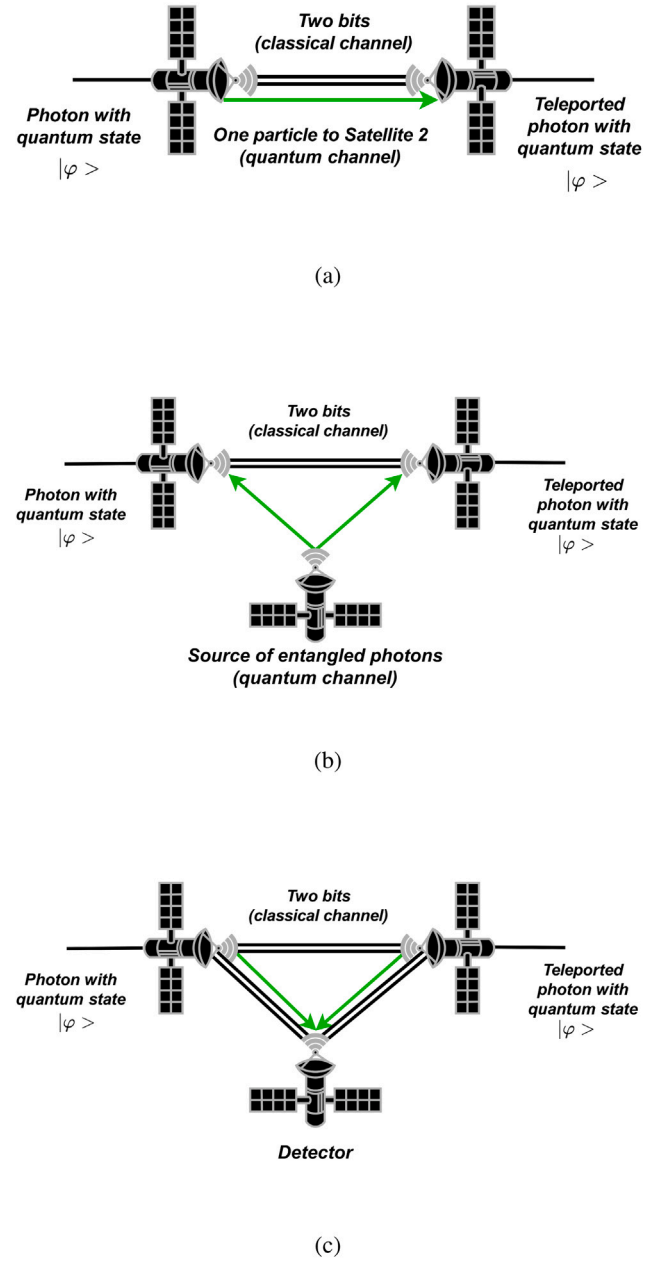

Fig. 5. In Fig. 5(a) the at-source strategy is depicted, whereas in Fig. 5(b) is represented the at-midpoint strategy. Finally, Fig. 5(c) represents the at-both-endpoints strategy.

A QR protocol requires some operations to establish the long-range Bell state, which can be used for quantum communication tasks such as entangled-based QKD or teleportation.

- Entanglement distribution: through this process is possible to create multiple entanglements among the links that connect network nodes.
- Entanglement purification: through this process is possible to increase the quality of the created entangled states starting from some with lower quality [88,89].
- Entanglement swapping: this procedure consists in a BSM performed within a node on two qubits that are halves of two independent EPR pairs. The BSM allows to create a longer entangled link connecting remote QRs.

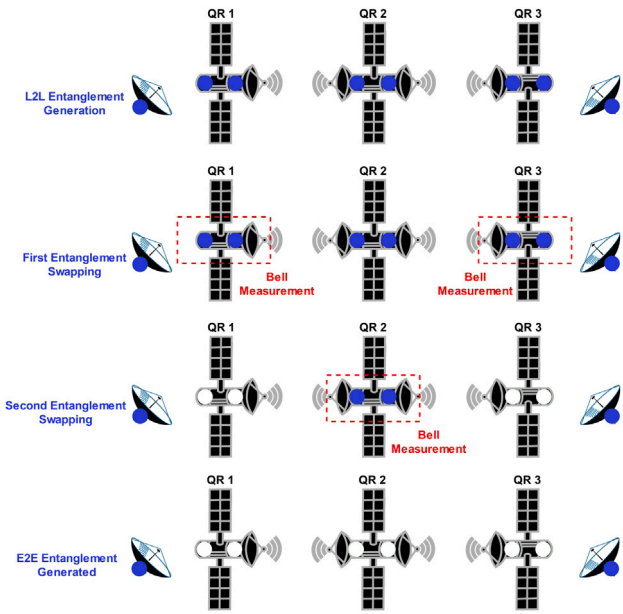

Fig. 6. Entanglement swapping between multiple satellites operating as QRs. The E2E entanglement generation starts with the Link-to-Link (L2L) entanglement generation between adjacent satellite repeaters. Therefore, the swapping operations are performed sequentially on multiple QRs until the E2E entanglement is obtained.

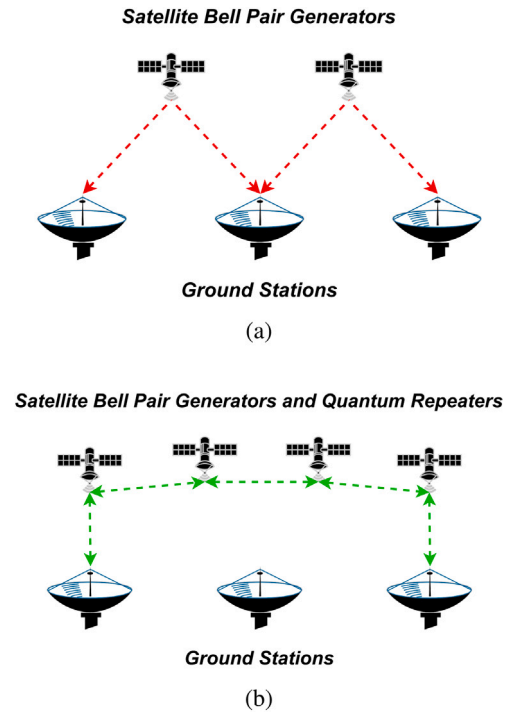

Fig. 7. In 7(a), each satellite is an entangled photon pair source, whereas in 7(b), entangled photon sources and QRs are placed on board of satellites [82,104].

Taking into consideration that the transmitted entangled states via a noisy channel do not result to be pure, the EPR pairs require a *purification* to be transformed to almost perfect entangled states. Measurements and local unitary operations are performed on the shared entangled pairs by the two actors, exchanging some classical messages in order to coordinate their actions, and sacrificing some of the entangled pairs to increase the purity of the remaining ones [90]. The resulted Bell pairs are almost perfectly pure and can be used together with classical messages in order to perform the teleportation of the unknown quantum states maintaining high fidelity [88,91]. In fact, as explained in [92], the deployment of a QR along the path can significantly augment the quantum teleportation fidelity.

As shown in Fig. 6, the E2E entanglement generation is obtained applying a specific QR protocol [93]. Specifically, multiple L2L entanglement are performed first between pairs of QMs of adjacent nodes via the transmission of photons entangled with their own memories [94,95] through a satellite quantum channel. The swapping is performed at the nodes that confirm the condition of entanglement with other QRs by receiving heralding signals from QRs positioned at long distances. Therefore, the involved device selects the pairs to be teleported and performs a BSM on the detained qubits of each EPR pair [96]. Hence, QMs must be entangled with photons for entanglement generation, and they have to preserve the entanglement at least until they receive heralding signals for entanglement swapping from remote nodes [64,97]. Thus, although two involved nodes never interacted with each other, after the swapping operation, their particles result entangled [98,99]. This protocol is typically applied to a sequence of several QRs performing multiple swapping procedures [100].

2.3. Quantum repeaters in non-terrestrial networks

As explained in Section 2.2, QRs are fundamental elements for the interconnection of multiple QCs over large distances. As reported in [101], QR chains have been created by interposing devices at short distances. In fact, as confirmed by [13], in which an attempt to optimize the network is made by proposing a routing algorithm suitably designed for QNs, the performance of OFs are acceptable only at distances of a few hundred kilometers. For these reasons, satellite quantum communications have been taken into consideration.

It is possible to infer from the studies conducted using a single satellite that:

- A satellite quantum link ensures better performance w.r.t. OFs.
- A satellite-to-satellite link allows to cover larger distances, however, it is necessary to perform further experiments [102].

Considering this, to create a global QN is necessary to take into consideration the integration of satellite links that allow reaching very long distances in combination with OF links for short and intermediate distances [103]. In [82] are examined two schemes, depicted in Fig. 7 for the realization of a QSN, which have been considered in the various experiments that are described in this survey. These schemes can be also used with all possible aerial platforms.

In the scheme depicted in Fig. 7(a), the satellites generate the Bell pairs and send them to the GSs. However, since satellites coverage is limited, it is impossible to realize a QN on a global scale with this configuration. In order to do this, some of the GSs must perform the function of QR in particular by performing the swapping operation to extend entanglement over greater distances. However, for the scheme adopted in Fig. 7(b), the satellites are equipped with both the hardware required to generate Bell pairs and the hardware required to perform the operations of a QR. This configuration allows interconnecting two QCs on the ground avoiding the terrestrial infrastructure, whereas its realization costs are higher.

As explained in [105], this type of communication can be performed using FSO technology. The commercialized FSO systems use Near-Infrared (NIR) and Short-Infrared (SIR) wavelength ranging respectively from 750 to 1450 nm and from 1400 to 3000 nm, and have already been used in OF communications. In particular, using these wavelengths, it is possible to create systems with a high data rate [106]. The FSO technology has already been considered for use on CubeSats deployed in LEO orbit [107] and also for communication towards other celestial bodies [108,109]. These satellites may employ

specific transmitters that operate at the 1550 nm wavelength [110], which is also ideal for quantum satellite communications [47,111,112]. Specifically, in [47] wavelengths in the NIR and SIR range equal to 850 nm and 1550 nm respectively for the quantum and classical channels have been considered, given their high efficiency. Moreover, a good level of miniaturization has been achieved, and in fact, in [113], it has been shown that it is possible to generate a quantum-secure key with a CubeSat. This result indicates that a QSN can be deployed with a limited number of launches, maintaining reasonable costs. As a matter of fact, considering that small quantum devices can be made [44], it is logical to assume that constellations of quantum CubeSats will be soon realized.

Several experiments and studies have been conducted by connecting GSs with individual satellites, and many frameworks for the analysis of constellations have been also developed. To this specific purpose, the application of SDN to control and manage the entanglement generation and swapping procedures on satellite constellations has also been taken into consideration. Furthermore, SDN paradigm has already been considered for the management of constellations composed of CubeSats on scenarios concerning classical networks [114].

2.4. Software-defined quantum networking

The SDN is a networking paradigm that allows for overcoming several limitations, such as difficulty of management and flexibility, which occur in current network infrastructures. Specifically, SDN allows separating the routing and forwarding decisions of networking devices from the DP. Hence, the network administration and management processes result to be simple, considering that the CP only manages the logical network topology data. The network traffic is orchestrated by the DP in line with the configuration established in the CP. The control operations are centralized in a Controller that manages the network policies [115,116] and performs direct control over the state in the DP elements through dedicated Application Programming Interface (API) [115], as depicted in Fig. 8.

According to this paradigm, the forwarding decisions that the DP made are flow-based, rather than destination-based. In particular, a flow is a group of packet field values and specific instructions treated with a matching criterion. The classification of the network traffic can be performed through the flow abstractions. Specifically, predefined matching rules are considered, and, therefore, the behavior of different network devices can be unified. The opportunity to program the flows enables significant versatility, constrained only to the capacities of the flow tables allocated in the forwarding apparatus. If flows are generated from entries that had been formerly installed in the DP device, the control traffic is not produced. All the packets that are part of the same known flow pass immediately through the switch, and a reduced delivery time is verified for them. If a switch receives a packet which is part of an unknown flow, the packet is forwarded to the Controller that inspects it and evaluates the action to be taken. Hence, the Controller instructs the switch to perform the same action for all the packets that are part of this particular flow [117].

SDN paradigm has also been considered in order to create Next-Generation Networks (NGNs), considering that it facilitates the integration between satellite and terrestrial networks [118]. As for terrestrial networks, in the satellite context, the SDN capabilities are managed by a centralized SDN Controller with a global view of the satellite network resources [119]. Specifically, the Controller can retrieve the data about the whole constellation from a file containing a Two-Line Element set (TLE), which is a data format that includes a list of orbital elements of multiple Earth-orbiting objects for a given point in time, the epoch [120]. Moreover, the Controller has a complete vision of the DP resources that are part of the satellite core network and the abstractions of the states of satellites to control the resources to offer, such as bandwidth in the access network. Thus, the satellite core network, which includes the network functionalities within the satellite gateway,

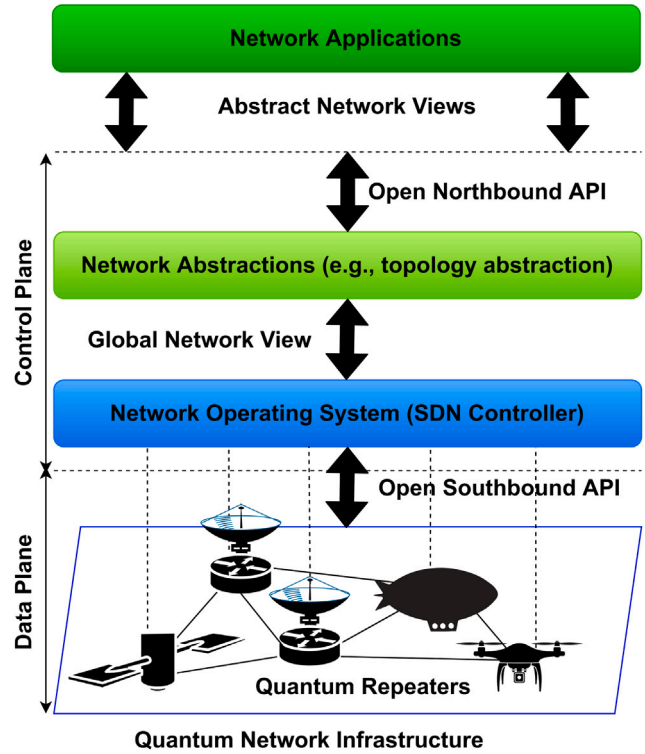

Fig. 8. Typical representation of a quantum SDN-based architecture for multiple purposes. The DP is composed of different platforms operating as QRs.

is managed by the SDN Controller that properly configures the flow rules of the QRs that connect the QCs located in proximity of the GSs, considering the priorities and policing if needed. However, the case of providing flexible and on-demand bandwidth in the satellite access subsystem is more challenging due to resource constraints and their statistical multiplexing [119].

Limited investigations have been conducted on using SDN in a QSN scenario. Given that the positions of LEO satellites change rapidly due to their high velocity, it is reasonable to take into consideration the SDN paradigm for the design of satellite quantum backbones. In a QSN, a Controller manages the global strategies for the dissemination of E2E EPR pairs. The Controller, after having managed the L2L entanglement generation procedure carries out the operations necessary to create the pairs on the E2E path by performing multiple entanglement swapping [101,121]. The control of a QN using SDN in order to manage specific quantum devices has been proposed, but however, many of the QKD protocols do not address control issues in true QNs aimed at E2E transmission of qubits, and long-lifetime entanglement generation [121]. Nevertheless, Section 3.2 describes some attempts aimed at the creation of E2E entanglement on a satellite path with dedicated protocols that have been made recently.

3. Non-terrestrial quantum systems: Deployments and experiments

This Section describes several practical experiments carried out in order to realize future quantum NTNs. The first ones are feasibility studies conducted to create QSNs and aimed at the creation of backbone controlled through SDN. Moreover, some of them are focused on HAPs considering specific channel models. Finally, this Section also includes studies related to QNs in a metropolitan context that integrate other types of aerial platforms, such as drones, also taking into consideration the use of SDN.

3.1. Feasibility studies on satellite quantum networks

Even though a constellation of quantum satellites has not been deployed yet, some models and frameworks to understand the current reachable performance and to provide some specifications for their design, have already been addressed. Besides the inter-satellite link analysis, the studies have focused on defining specific architectures.

One of the first studies to demonstrate the feasibility of quantum satellite communications was [122]. Specifically, a 144 km link was created between two GSs, which is longer than some satellite-to-ground links, thereby demonstrating the feasibility of both FSO terrestrial and satellite links for QKD applications.

In [103] a rate calculation for a repeater architecture described in Fig. 7(a) has been performed. In this scheme, many satellites disseminate Bell pairs between the GSs, and the entanglement can be extended by performing the entanglement swapping on the GSs themselves [123]. Specifically, the cumulative entanglement rate obtained during an entire day as the distance between GSs varies from 4000 to 20 000 km taking into consideration from 4 to 8 inter-satellite links has been evaluated. Moreover, the entanglement rate at a distance of 20 000 km has been estimated as the memory features of the involved devices change, always taking into account several links and a time window of one day. The entanglement rate has also been evaluated varying the *Quantum Nondemolition Detectors (QND) efficiency* in similar operational conditions. A QND detects the presence of a photon in a non-destructive manner [124], and the more the value of QND efficiency approaches the value 1, the higher the entanglement rate is. As explained in [125], the objective is to implement measurement schemes in which feedback noise is not coupled to the quantity of interest, which results are not affected by the measurement process; therefore, it would be possible to conduct repeated measurement operations with extremely high precision.

In [104], different approaches are investigated for global quantum networking considering space and ground networks scenarios. To create a connection between the two involved GSs, the study considers satellites equipped with QMs, operating as QRs. Specifically, measurement-device independent QKD protocols have been implemented to guarantee high rates and device-independent security onboard satellites in a LOS setting. The study confirms that quantum satellite repeaters allow for achieving better performance in terms of entanglement rates on a planetary scale than OFs solutions. Specifically, the study analized a Vienna-Sydney link separated by numerous GSs by varying the meteorologic conditions and considering different atmospheric effects. Findings indicate that on the basis of the considered effects, the solution depicted in Fig. 7(a) is impractical w.r.t. the solution shown in Fig. 7(b). The impact of losses using multiple QMs characterized by different efficiency values is also investigated in the paper.

An attempt to provide some guidelines for the creation of a QSN was made in [126]. The study states that the prospective global QN could be a composition of networks of LEO satellites, which will be connected to a network of satellites deployed in both Medium Earth Orbit (MEO) and Geostationary Earth Orbit (GEO). In particular, this network can play a significant role in scenarios related to the Internet of Things (IoT) architectures, which are composed of multiple distributed nodes interconnected through different kinds of channels to enable new functionalities in communication, computing, and sensing and necessitate high security that can be reached by using specific QKD protocols. For instance, the backbone could be used to secure multiple IoT domains distributed on a global scale.

These kinds of architectures, including the use of several constellations, have also been evaluated in different papers. For instance, in [47] a double-layer QSN architecture has been proposed. Specifically, the paper proposes an architecture that consists of GEO and LEO satellites and a GEO-LEO Routing and Key Allocation algorithm conceived to distribute the keys on an E2E basis. The results were obtained on a scenario related to a network consisting of 66 LEO satellites, 3 GEO, and 25 GSs placed on the surface of Earth.

3.2. Software-defined networking satellite backbones

The studies that have been explained in Section 3.1 do not consider the use of SDN paradigm, which as described in [101], proves to be very useful in the management of QNs. In fact, as mentioned in Section 2.4, the SDN Controller, thanks to its complete control of the network, can manage the creation of the Bell pairs in an effective manner on the entire E2E path [71].

Considering that quantum satellites can be used to generate secret keys for terrestrial nodes, in [127] is proposed a network architecture managed by SDN Controllers on the ground with different roles, which are interfaced to the LEO satellite network composed of quantum satellites through a network of GEO satellites in order to effectively distribute quantum keys. In particular, the study proved the performance of multiple QKD constellations on which it was applied a dedicated Routing and Resource Allocation in Free Space QKD (RRA-FSQKD) algorithm. The obtained findings indicate that the performance of the constellation in providing key relay service is related to the condition of an inter-satellite link. Moreover, the key generation probability of success increases considering the existence of satellites deployed in different orbital planes. If the involved satellites are part of the same plane, the satellites disposed in distinct orbits cannot accomplish the key dissemination; therefore, the number of satellite orbits impacts significantly on system performance.

Nevertheless, to enable other features of the QI, such as distributed computing [128] between multiple QCs even thousands of kilometers apart, it is necessary to perform further studies also considering different architectures. For instance, in [7] in addition to evaluating the maximum achievable rate on a single link in the space vacuum, different path selection algorithms have been studied on a satellite constellation composed of LEO satellites based on the existing IRIDIUM-Next constellation. In particular, two distributed algorithms and a centralized one have been considered. The distributed algorithms are a Modified Random Walk (MRW) and an Ant Colony Optimization (ACO), while the centralized one is a classical version of Dijkstra's algorithm. The MRW algorithm is the simplest, and it is based on choosing the next closest satellite, avoiding satellites already visited. The ACO algorithm instead is based on the behavior of social insect colonies like ants [129]. An insect that encounters an already crossed trail may detect it and decide to cross it with a high probability. Furthermore, the trail intensity is reinforced by a variable proportional to a decay factor and the number of packets in transit on the involved nodes during a specific time interval [130]. Finally, the classical Dijkstra's algorithm requires a centralized routing CP and this can be achieved employing the SDN paradigm. The performance of the considered algorithms have been evaluated in terms of entanglement rate, also considering the link length, the number of hops, and the distance between the GSs. The paper also provides some architectural elements, such as a logical scheme of the functionalities that a GS, which coordinates the quantum constellation must perform, as depicted in Fig. 9.

Furthermore, always considering the architecture designed in [7], in [8] the performances of different constellations have been analyzed. Three constellations, one MEO and two LEO, have been investigated, also considering the duration of the best paths that are calculated by the SDN Controller. Constellations performance was evaluated by varying the distance between GSs by examining the achievable rate based on total path distance. As can be seen from this paper, a MEO constellation does not allow to obtain satisfactory performance considering that the achievable entanglement rate values are less than 2 Bell pairs/s. In particular, it is inferred that a denser and lower constellation allows for better performance. Furthermore, the average duration of the path has been calculated, which is useful information for the SDN Controller to better manage the path. This data is useful to configure the devices along the path to operate in a specific time interval that corresponds to the duration of the path, as was done in [9] explained in the following.

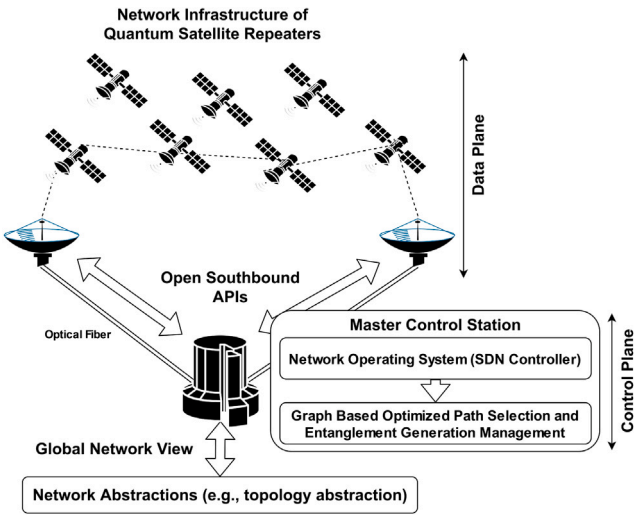

Fig. 9. Quantum SDN-based satellite architecture adopted in [7,8].

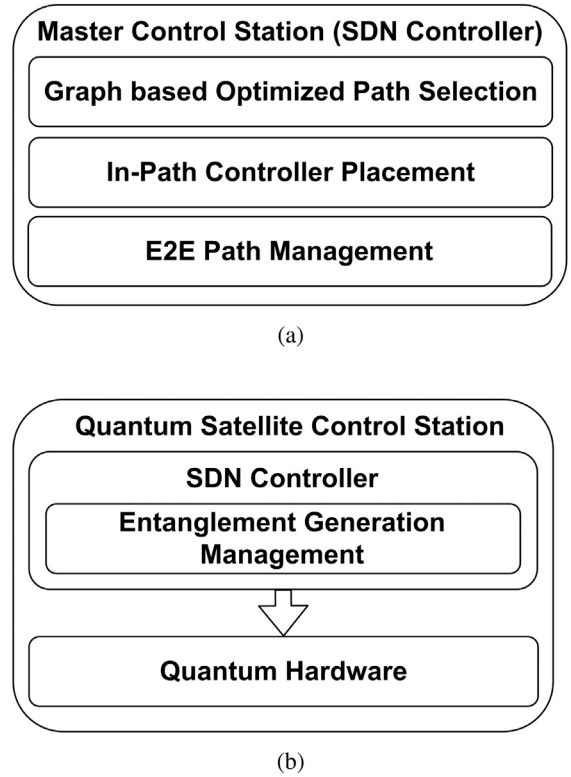

Fig. 10. Logical diagram showing the features of the elements described in [9]. In 10(a) the features of the Controller deployed in the MCS are highlighted, whereas in 10(b) the functionalities of the Controllers embedded in the constellation are described.

Moreover, [131] provides some guidelines for designing an effective QSB that makes use of the SDN paradigm, with emphasis on the CP procedures and interfaces. The paper compares the E2E entanglement generation procedures depicted in Figs. 5(a) and 5(b) on a single Controller scenario. Subsequently, the most effective procedure is applied to two different scenarios: in the first it is employed a single Controller on Earth capable of managing the entire constellation and in the second multiple Controllers are allocated within the constellation. According to the study, the adoption of an architecture that includes multiple mobile Controllers allows to solve many issues and achieves better performance than an architecture with a single Controller on Earth. Besides limiting the traffic on the Satellite-to-Ground links that are particularly critical, it can allow reducing the overall delay, which is crucial in distributed QCing applications. The study evaluates the performance considering the overhead and latency required to establish an E2E entanglement. Furthermore, the proposed SDN backbone, that uses several mobile Controllers deployed in the constellation, achieves superior performance in terms of both the considered parameters concerning the architecture with a single Controller on Earth.

In [9], an alternative scheme is considered, consisting of a Master Control Station (MCS) on the ground with the task of coordinating the constellation and calculate the best path. Moreover, the MCS allocates some SDN Controllers in the constellation itself with the functionality to manage the procedures of entanglement generation within their domains. In fact, as explained in [132], different approaches can be used when designing the CP of a network based on SDN. The CP is therefore composed of two types of Controllers that operate in a coordinated fashion and whose functionalities are expressed in Fig. 10. The paper also defines a possible control packet format and a Network Layer protocol. The protocol, which is specifically designed for the proposed architecture, is organized in two main phases:

1. Setup of the entire satellite path considering the requests from the GSs.
2. E2E entanglement generation on the configured satellite path.

In the first phase of the protocol, the MCS processes a request from a GS willing to create an entangled link with another GS. Then, the MCS starts a Graph based Optimized Path Selection procedure to calculate the best routes and their durations. Meanwhile, it interacts with the selected satellites and activates the control processes embedded in the constellation. During the second phase, the Controllers embedded in the constellation manage the operations of entanglement generation on the entire path. The operations of entanglement generation occur throughout the whole duration of the path. When it is no longer the optimum one, the procedure restarts, and the resources are allocated on a new E2E path. The whole procedure is repeated until the end of the communication.

A hierarchical architecture was also proposed subsequently in [71]. Specifically, the architecture consists of three levels of Controllers with different roles. If a domain receives a user request that is unable to manage by itself, e.g., a connection with a user located in another domain, it relies on a higher-level Controller that manages multiple domains. The request reaches a common Controller for both users that manages the connection, defines the optimal E2E path on which to perform entanglement swapping, and sends it back to the lower level Controllers to be fulfilled. The paper provides some preliminary tests performed with single satellites considering the most common wavelengths used for quantum communications.

In [133], an SDN-based architecture for quantum key exchange via a satellite constellation consisting of GEO and LEO satellites is proposed. Specifically, the architecture includes a Quantum Key Pool (QKP) layer, which allows to save the secret keys for timely provisioning. The keys are obtained from the Physical Layer and provided to the QKP layer for storage to reduce the key wastage. The SDN Controller includes a QKP layer control module and a Physical Layer control module. The QKP control module routes the requests over the QKP layer and manages the rest of the keys for provisioning, while the Physical Layer module controls the allocation and deallocation of the links between the satellites and the associated GSs.

In [134] a Temporal Networks (TNs) optimization strategy embedded in an SDN Controller ecosystem is proposed. Specifically, the path durations are dynamically optimized in order to maximize the entanglement rate, while limiting the satellite to ground and inter-satellite disconnections.

The QSBs can enhance existing applications and enable new ones. Some possible uses of the QI can already be identified and classified [135]. Specifically, Section 4 outlines the applications enabled by the explained architectures, and the related experiments.

3.3. Feasibility studies on high altitude platforms

As explained in [37], HAPs can provide a cost-effective solution for broadband data access in areas where terrestrial infrastructures are limited. As a matter of fact, the interconnection of multiple airships deployed in the stratosphere can ensure satisfactory performance considering that they operate above the clouds and the inter-HAPs link is not significantly perturbed [136,137]. However, not many studies have yet been conducted regarding the use of these platforms in quantum communications. A significant investigation is reported in [138], in which an estimate of the link budget is made considering the Kruse's model [139,140] for optical communications in the lower atmosphere and beam wandering problems. Specifically, the channel loss is estimated under different meteorological conditions, and the Quantum Bit Error Rate (QBER) estimation is made by also varying the angle of incidence w.r.t. the GS.

The conducted studies only address the feasibility of QKD applications, but the use of HAPs for other purposes, such as interconnecting devices for distributed computing and integrating them with other communication devices, is certainly worth considering.

3.4. Feasibility studies on quantum drone networks

Other than considering satellite networks controlled through SDN [101], it is necessary to evaluate different communication technologies [141] and diverse kinds of aerial platforms such as drones [142]. In fact, a network composed of multiple Quantum Drones (QDs) is solution that can be disposed easily on-demand and that allows covering limited areas where an ad hoc quantum computation needs to be implemented. The use of a swarm of QDs also allows to overcome the static deployment typical of OFs and related problems.

Many studies regard the use of QDs to verify the feasibility of several QKD protocols on individual drones; however, they do not consider a properly designed CP. In fact, the SDN paradigm has been applied only in a few studies. For instance, in [143], the BB84 QKD protocol [144] has been tested on a simple scenario composed of a unique drone connected to a single GS. Specifically, they measured the QBER sending multiple Qubit strings with different lengths by varying the distance between the GS and the drone.

In [145], a quantum cryptographic network that uses a couple of multi-copter drones has been studied. In particular, it focuses on developing an optical payload for QKD with the capability to maintain the correct pointing between two drones. Furthermore, the air-to-air signal coupling between a couple of drones has been evaluated in [146].

Moreover, in [141], a free-space quantum communication through drones has been analyzed. Specifically, the scenario consists of a single quantum drone equipped with an entangled photon generator, which relays Bell pairs to a second drone that delivers them to the destination. The paper reports the correlation curves measured in the laboratory, the tracking errors of transceivers in the drone-to-drone link and drone-to-ground links, and finally, the correlation function results obtained during the experiment for the Clauser–Horne–Shimony–Holt (CHSH) inequality [147,148].

In addition, the quality of the entangled states generated onboard a single drone that connected two GSs has been verified in various daylight and meteorological conditions in [149]. Specifically, the paper reports the diffraction loss considering different lens diameters and by varying the distance between the drone and the GS. Also, in this case, the values obtained for CHSH inequality were verified but during different times of the day and under specific meteorological conditions. A contribution that examines the use of SDN in the context of drone networks is [63]. Initially, the study evaluates the fidelity obtained on different E2E paths as the wavelength varies and, thus, evaluates the technology that can be employed. Furthermore, the performance is evaluated considering different meteorological conditions based on Kruse's model [139,140] and by varying the distance between the connected GSs. The study provides also essential architectural specifications for designing a network of QDs manageable through SDN and defines a dedicated SDN-based protocol with performance evaluation. Specifically, the topology can be optimized and controlled considering an objective function that takes into account multiple QoS parameters, which allows for optimal mission planning. The architecture proposed in the study, which enables the Bell pairs generation on an E2E basis, is suitable for distributed QCing and entanglement-based QKD applications.

Finally, in [150], an algorithm based on quantum multi-agent reinforcement learning dedicated to the autonomous construction of a robust mobile access system using multiple QDs has been presented. In particular, the paper evaluates the performance of the proposed algorithm considering the training speed, wireless service quality, and network reliability in several scenarios.

4. Quantum applications

This Section reviews several studies on possible applications that could be globally scaled through a properly controlled NTN. Specifically, some studies on QKD and on distributed computing architectures are examined. Finally, the Section considers studies on QPSs.

4.1. Terrestrial networks and satellite QKD experiments

Security has been an important area in research, engineering, and operations for the classical Internet [48]. The existing online services store or transfer sensible data that can be a primary target for hackers. Furthermore, many organizations and companies can also be affected by cyber-attacks that can cause significant economic losses [151]. In order to deal with these issues, a classical network security system typically relies on multiple layers of protection and consists of different components, including networking monitoring and security software along with specific hardware and appliances [152]. However, the progress of QCing also require research into new technologies that can enhance network security. As a matter of fact, a QC can solve the prime-number factorization problem exponentially faster than a classical computer [153]. Even if QCing is not yet a consolidated practice, some experiments have successfully been conducted in several laboratories [154]. In order to address these issues, Quantum cryptography has been considered. As reported in [155], many terrestrial QNs have been recently implemented and for several of them is planned a future integration with free-space networks. Among the 27 networks that are mentioned in [155] are the Madrid QKD metro network based on Huawei's QKD systems; the Cambridge QKD metro network using Toshiba's QKD systems; and, finally, the Cambridge-Ipswich and the Bristol QKD backbone networks that rely on ID Quantique QKD systems; the Hefei QKD metro network relying on QuantumCTek QKD systems. Moreover, several organizations are developing specific standards to enhance system interoperability. For instance, ETSI is addressing control interfaces for QKD networks managed via SDN in [156].

The terrestrial networks that are discussed in [155] are based on different technologies that have already been also tested in the free-space and satellite scenario [157]. As explained in [153], many tests have been carried out for satellite QKD, considering the reasons explained in Section 2.1, and many studies are focused on the realization of photon sources dedicated to increasingly efficient QKD applications [158].

In [159] an experiment that emulates a photon source onboard a single satellite is described. In particular, they used the telescope of an observatory in Matera under the control of the Italian Space

Agency to identify transmitted photons and demonstrate the possibility to establish a quantum channel between a single satellite and a GS. The LEO Ajisai satellite, has been utilized to emulate a photonic source onboard a satellite using the reflection of laser pulses from a Satellite Laser Ranging (SLR). Specifically, the research group measured the number of detected photons per second, that is, the Detector Count Rate (DCR). The study is a first step towards effective measurements of single photons, which is fundamental to establishing quantum communication.

Furthermore, in [69] an Italian research group has demonstrated that it is possible to preserve the polarization of a single photon over a satellite-to-ground channel. Two single-photon detectors have been used to measure two different polarizations, and they derived the QBER from the obtained data. The study considered four different polarization input states and analyzed four specific 10 s intervals. In particular, it shows the corresponding histograms related to the counts at the receiver for each single-photon detector w.r.t. the detection time. Moreover, the study reports for each of the four considered satellites the measured QBER and the QBER calculated after the background subtraction and demonstrates the feasibility of the BB84 protocol on a single satellite scenario, requiring an average QBER below 11%. The study also characterizes the detection rates and link budgets for the involved satellites.

The deployment of the LEO Micius satellite in 2016, orbiting at 500 km altitude, paved the way to many experiments. Five GSs for optical communications have been built in different locations in China in order to perform the experiments [160]. As described in [70], in a joint experiment involving Chinese and Austrian researchers a quantum communication between multiple locations on Earth at a continental scale distance of maximum 7600 km has been demonstrated. In particular, during the experiments, some images were transmitted and was also held a 1 h and 15 min video conference.

As reported in [111], the key rate that can be achieved through the Micius satellite in a satellite to ground communication considering a distance of 1200 km is of the order of kHz, which is about 20 times the order of magnitude achievable with OFs. This paper assesses the sifted key rate and the QBER during time w.r.t. the distance between the GS and the satellite. The efficiencies of the satellite links have been obtained calculating the ratio between the intensity of the photons received by the GS detector with that sensed at the transmitter that is a photonic source onboard the satellite. The outcomes confirmed the higher performance of the satellite link w.r.t. an OF link and is coherent with other reported data.

In the experiment described in [161], they have demonstrated the distribution of two entangled photons from a satellite to two GSs at a distance of 1203 km, considering the scheme described in Fig. 7(a), and they have observed the preservation of the entangled particles. This result is significant because it demonstrates the viability in implementing entangled-based QKD protocols such as the E91, which is described in [148].

In [162], the feasibility of a satellite ground link through the use of Continuous-Variable QKD (CVQKD) technology has been analyzed. The research group derived the probability distribution of the transmission efficiency related to the quantum channel in multiple cases and they extrapolated the secret key rate. Given the channel fluctuations that characterize satellite communications, the paper proposes an approach of data analysis that takes into account orbit subdivision. The results revealed that coherent state modulation and detection is a feasible solution in the case of LEO satellites quantum communication. The communication with higher orbits, may be influenced by finite-size effects if the transmission rate remains low or if the partition of the orbit is not optimal. However, by merging several satellite passes, the communication range could be extended beyond 2000 km. This study is significant considering that CVQKD offers many advantages w.r.t. Discrete-Variable QKD (DVQKD). In particular, CVQKD guarantees higher secure key rate and cost-effective detection using homodyne receivers, and it is compatible with current classical communication technologies. Moreover, this approach is efficient even in daylight, considering that homodyne detectors offer significant robustness to background light sources that negatively affect single-photon systems [163].

Considering that atmospheric phenomena such as clouds, fog, and haze have a significant impact on quantum communication and key generation, in [164], a specific method was applied to mitigate these effects. Specifically, they used a genetic algorithm wave-front shaping method that gathers the transmitted photons within a predefined area and maintains the spot shape as possible similar to the Gaussian distribution. The study states that it was possible to increase the transmission efficiency of the BB84 protocol by 250 times compared to the common analyzed cases, which do not consider this type of compensation.

An attempt at constellation design to support quantum cryptography was made in [165]. Specifically, the study considers global and regional constellations, evaluating the key rate on a daily basis. Moreover, in [166], the key rate by considering two scenarios related to satellite communications between two GSs placed at different distances, taking into account various path loss issues have been derived. In particular, in the first scenario, only a few satellites were equipped with QMs. Therefore, only these satellites are capable of storing qubits and performing swapping operations. Conversely, in the second case, swapping operations are performed on all satellites in the E2E path, which in fact are equipped with the same quantum hardware.

In [167], the accesses selection and inter-satellite routing issues considering a LEO constellation designed for QKD are studied. Specifically, the paper proposes an access selection and a dedicated routing algorithm that are tested on a constellation composed of 9 orbital planes and 40 GSs located in 40 different cities on Earth. The BB84 protocol was applied on the given scenario.

Despite the experiments conducted until now demonstrate the feasibility of satellite QKD, many issues are still open. Given that the QR systems are also made of classical components and interfaces, the most critical issues concern the classical elements that compose these systems. Specific attacks that involve quantum and classical parts must be considered for designing new architectures as robust as possible [48].

4.2. Distributed computing and quantum cloud

Among the new applications that the QI is likely to enable, it is important to consider distributed quantum computing and novel quantum cloud systems. Considering that the involved processors rely on different physical principles and technologies, several research groups are working on an extensive characterization of the QI protocol stack [168].

Taking into consideration the recent evolution of QCs, it is reasonable that the security of the current Cloud systems results compromised. In fact, given their high computational capacity, these systems are able to easily bypass current security measures [169]. Therefore, there is a necessity to design a new Cloud concept that includes QCs and quantum communication technologies to address any potential attack.

In [170], the constituent elements of a Quantum Cloud (QCloud) system are described. Specifically, it is composed of the following elements:

- The quantum hardware is composed of qubits, which can be manufactured with different technologies, and the related hardware.
- The control system is composed of the necessary elements to operate on the quantum hardware, guiding its evolution and retrieving the outcomes.
- The executor allows to orchestrate the control system to execute quantum programs and returns the results of the measurements to the user.
- The compiler produces instrument binaries for the executor considering the quantum programs.

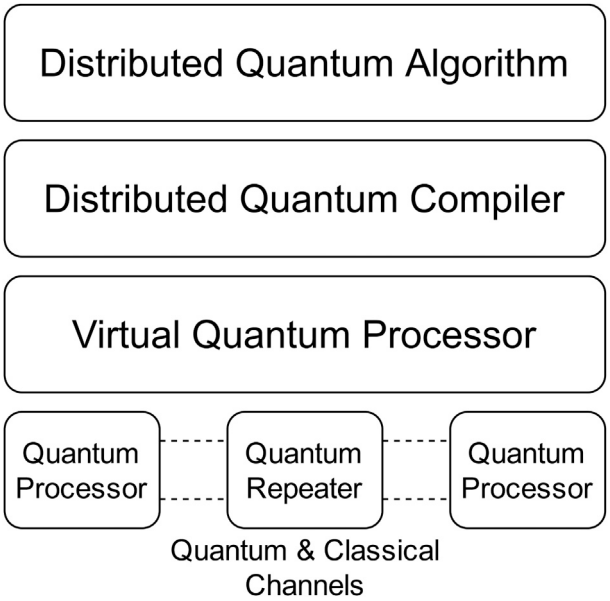

Fig. 11. Logical scheme of the distributed QCing ecosystem described in [172].

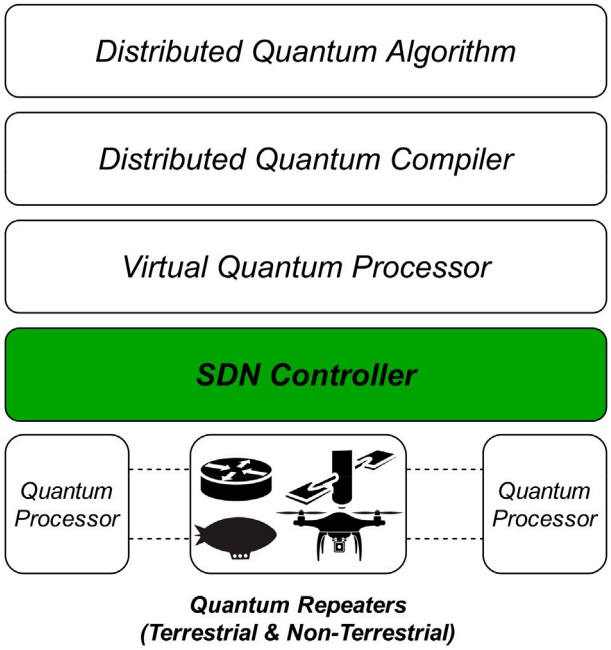

Fig. 12. In order to perform large-scale QCing operations, at least one SDN Controller and QRs installed on aerial platforms must be integrated.

As explained in [128,171], a network of quantum nodes which comprises k nodes each with n quantum bits has a state space of dimension 2^{kn}. Therefore, considering k remote QCs, that dedicate no less than one qubit for the teleporting process, it is possible to obtain a virtual quantum device consisting of up to $kn - k$ qubits. The operations must be properly orchestrated between the single devices and considering this, an initial architectural scheme for distributed computing has been provided in [172]. As depicted in Fig. 11, the lowest layer consists of the classical and quantum channels, QRs, and quantum processors. The Virtual Quantum Processor (VQP) is the level immediately above, acting as an interface for the Distributed Quantum Compiler (DQP). The VQP optimizes the underlying circuit in such a way that the DQP can operate regardless of the kind of network. Specifically, the DQP maps a quantum algorithm in a series of operations, which can be executed locally and remotely, optimizing the available computational resources taking into account the constraints of the network. The DQP could interface with an SDN Controller that performs the operations described in some experiments shown in the previous Section [7–9]. This peculiarity could be essential, particularly in the case of a time-varying network, such as a satellite scenario. Furthermore, with the control reached using SDN, the paths can be optimized to improve the management of the entanglement generation between remote QCs to make full and efficiently use the space of states obtainable combining multiple quantum devices. In fact, as illustrated in [49], the ultimate goal is to interconnect clusters of QCs, i.e. different Quantum Data Centers (QDCs). Nevertheless, to reach this target, it is required complete control of the heterogeneous network that could include both terrestrial and satellite segments and the different technologies used to manufacture the classical and quantum hardware. In particular, the interconnection of several QCs that work in a coordinated manner in an urban area could be realized through existing OF links; however, the interconnection of multiple geographically distributed QDCs could be performed with the use of the architectures discussed in Section 3.2 utilizing the features of SDN.

Other studies have been done in an attempt to analyze and define a QCloud architecture [173]. Many of them are aimed at using quantum technologies to secure Cloud systems [174]. The designed model is conceived to authenticate the users in the Cloud with the use of quantum cryptography. In particular, the study analyzed the performance of the BB84 protocol to set up a secure quantum channel.

Although a definitive architecture for a QCloud system does not yet exist, in [175] a QN security framework is proposed. The study introduces a Quantum Key model application specifically designed to increase system security, introducing a dedicated secure key layer, acting as an intermediary between the host kernel and containers. Moreover, the use of quantum keys can enhance QCloud systems' security. In particular, specific control levels are established among containers and container daemons to create ad hoc network links between servers deployed in the same or in distinct Data Centers. Once a container is initialized, it has limited privileges by default. As a consequence, the container and the related applications have access restrictions w.r.t. the resources in the host kernel. Nevertheless, if an application requires root privileges, a classical message is sent via a channel that connects the Host Server (HS) and the Quantum Server (QS). A series of requests are sent by the Daemon to the QS involving multiple applications such as routing, file modifications, and networking. The system considered in [175] works as follows:

1. A QS includes an EPR source, which generates Bell pairs.
2. The QS and the HS exchange Bell pairs between each other.
3. The Bell pair particle detained by the QS interacts with the informative qubit.
4. The quantum information qubits are extrapolated by Quantum Encryptors converting the key data.
5. The generated key is used to set up a secure communication.
6. The root privileges are attributed to those containers whose keys have been created with success.

The paper also asserts that additional algorithms need to be developed to ensure an effective update of the QN topology and routing tables, also taking into consideration the distributed Bell pairs amongst QCs. Moreover, it is also required to develop specific purification protocols to also take into account other issues such as fragility of the entangled states, the quantum noise [176], and decoherence typically associated with quantum data.

The described problems can be easily addressed by considering the use of SDN paradigm applied to quantum NTNs, as depicted in Fig. 12, where an SDN Controller is introduced w.r.t. the architecture

in Fig. 11. In fact, with an SDN Controller, it is possible to provide an efficient update of the QN topology, and routing tables, also providing an efficient distribution of the EPR pairs among QCs and QRs installed in different terrestrial and NT platforms with the opportunity to extend these applications on a global scale. Furthermore, the introduction of this element is crucial considering the variability of the scenario and the mobility of many elements within specific time intervals. In this manner, it is possible to achieve specific performance in terms of fidelity and entanglement rate while minimizing the number of required elements and consequently the overhead.

4.3. Synchronization and positioning

An attempt to design a quantum synchronization satellite system for multiple applications that require microsecond accuracy, such as financial transactions [177] or specific industrial processes, was made in [178]. Specifically, the paper simulated the spectral density of resonators in a three-qubit satellite system using lasers operating at a wavelength of 813.32 nm and obtaining a clock transition frequency of 262.976 THz.

Moreover, clock synchronization is significant also in quantum communication applications, considering that precise synchronization between a couple of agents is required to correctly associate the symbols transmitted and in order to reduce the QBER [41]. As a matter of fact, a synchronization of the order of ns between two terminals is required to fully exploit the high timing accuracy of current single photon detectors, which can be reached by sending periodic reference pulses between the two terminals [111].

The most worthwhile application for the civilian and military sectors is navigation, considering the degree of accuracy that can be achieved in applications as for instance the Global Positioning System (GPS). In [52], a methodology that maximizes the performance of a network composed of multiple clocks with the support of quantum technologies has been considered. The strength of this approach is the opportunity to leverage all the resources on each node. Distributed networks consisting of quantum-limited clocks very distant from each other, for instance, in the case of satellite-based clocks, can operate as global clocks. Specifically, all members share their own resources that are combined in a quantum coherent mode achieving greater clock stability and distributing the temporal data in real-time. To obtain this outcome, it is required a combination of modern atomic clock apparatuses, using highly phase-coherent lasers with quantum technologies, to connect distant quantum devices in a QN. This approach has already been considered in single atomic clocks to improve their performance [179]. By increasing the number of parties, it is possible to increase the stability of each involved party's local clock signal without compromising security. The application of quantum communication techniques allows achieving the highest possible system security level, taking into account that only the involved parties takes the benefits of such a temporal accuracy. Specifically, a set of n atomic clocks onboard the nodes composing the network are involved. In particular, the nodes are composed of numerous atoms, that are, clock qubits, operating at a reference frequency for each distinct geographical zones, and each clock has its independent Local Oscillator (LO). A periodic query of the qubits is performed to maintain the synchronization, and furthermore, the measured data are also utilized to stabilize the LO at the atomic transition reference frequency. A number of qubits are made available to create entangled states extending to all the involved nodes. Furthermore, as shown in Fig. 13, the obtained data is sent to a specific Central Node (CN), which is utilized as a center-of-mass node to stabilize the multiple involved LOs. The entangled network states provide extraordinarily precise data about the center-of-mass deviation of all LOs from the atomic resonance. Thus, a cycle is composed of the following stages:

- Initialization: the state of the atomic clock is prepaired.

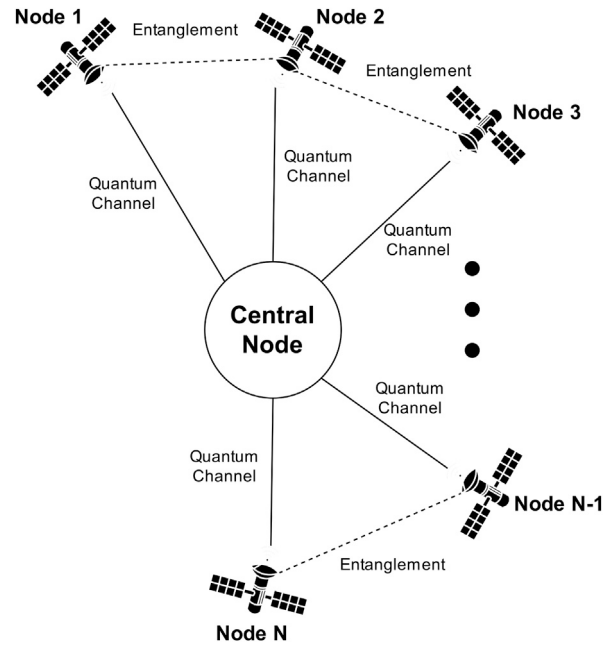

Fig. 13. Schematic representation of the satellite network proposed in [52]. The nodes interrogate their LOs and the obtained information is sent to a CN. Therefore, a remarkably accurate clock signal for all users is obtained.

- Measurement: query by the LOs.
- Feedback: adjustment of the laser frequency on the basis of the measurement outcome.

After several cycles, the LOs of every single node acquire accuracy and stability, resulting from the interrogation of atoms in the whole network.

The proposed QN can also be used as a large-scale quantum sensor to investigate the fundamental laws of physics, permitting us to understand better the relativity and the laws that govern space–time and quantum physics.

The QPS, for the first time proposed in [53], can significantly improve the position accuracy of the traditional GNSS as required by the new location applications. The GNSSs offer several location services in order to obtain the exact coordinates and conduct the navigation of vehicles suitable for a wide variety of activities and scenarios by utilizing a specifically designed constellation providing signals that transmit data related to the position and time to GNSS receivers. The first system to be deployed was GPS by the US Department of Defense for military operations and then released to civilian users, which uses a combination of data provided by multiple satellites. In particular, the vehicles equipped with a GPS receiver measure the time of arrival of the multiple signals received from four satellites and are able to compute their coordinates in an automatic way. Although the wide coverage of GPS, the position data cannot be provided continuously in consequence of the blockage of signals, for instance, in metropolitan areas and with a maximum accuracy of approximately 5 m. Furthermore, security issues can arise considering that the signal can be overridden or eavesdropped. The QPS enhances the accuracy by a factor $\sqrt{\eta\kappa}$, where η is the number of frequency-entangled pulses and the factor $\sqrt{\kappa}$ is obtained by employing squeezed pulses of κ quanta [180,181]. Furthermore, considering that these systems make use of quantum technologies their security is ensured by the no-cloning theorem.

An interesting review on QPS can be found in [182], which introduces the main concepts of a quantum navigation system dealing with the aspects of quantum navigation classification, core technology, and development trends. To realize a QPS, optical communication technologies are necessary, along with acquisition, tracking, and aiming

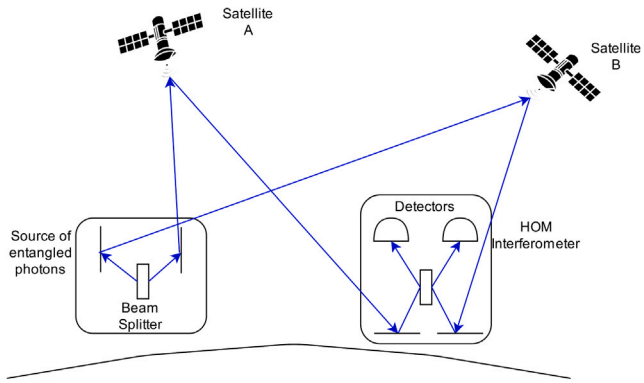

Fig. 14. Diagram that represents a QPS.

techniques and, therefore, the apparatus typically employed in classical satellite laser communications. The positioning and the processed clock synchronization are independent in a QPS. The temporal disparity between the clock detained by the user and the clock of the system that is positioned in proximity to the origin of the coordinate system is measured with high accuracy considering the second-order quantum coherence. The clock detained by the user is synchronized with the system clock, and this synchronization requires only a short-term stability, given that the two-photon coincidence count measurement of the Hong-on-Mandel (HOM) interferometer [183,184] only requires the clock to remain stable for a limited measurement period.

Moreover, some guidelines for designing a QPS are described in [185]. The active QPS uses satellites as sources for the transmission and reception of quantum signals, and it is composed of a ground unit, the satellite system, and users. The QPS architecture requires a couple of LEO satellites for each direction of a perpendicular coordinate system with the origin in the center of Earth. A photonic source transmits towards both satellites, reflecting the signal to the user. Setting the optical delay unit properly, the couple of photons can get the user's interferometer simultaneously. Ultimately, knowing the distance between the satellites and the delay, the user position can be determined [182,185].

The realization of a QPS is primarily influenced by the preparation of the entangled photons. In addition, the detection of photons needs to be efficient with an accuracy close to 100%. Moreover, the atmospheric attenuation has a significant effect within the first 10 km of altitude, influencing the position of users on the ground; nevertheless, the spacecraft position is not so relevant. Furthermore, a QPS requires that the stability of HOM interferometer is maintained for a limited measurement time, whereas for traditional localization systems are required longer periods (see Fig. 14).

5. Conclusions

This survey, as well as providing some basic physics concepts on quantum technology, mainly investigating the role of satellites and various types of aerial platforms in quantum communications towards the global QI vision. As a matter of fact, considering that quantum states are very difficult to preserve because they are subject to decoherence, and cannot be sent through OFs without including a large number of QRs, a quantum NT backbone could offer an efficient way to transfer quantum information across multiple quantum devices on the ground.

Specifically, in the first part, fundamental concepts such as the basic unit of information, the concept of entanglement and quantum teleportation, and the phenomenon of decoherence due to the channel and the devices themselves on the qubits and negatively affects the Bell pairs created are explained. In addition, some basic concepts of QKD are provided, as also the related metrics.

Furthermore, the paper explains the motivations related to the implementation of quantum NTNs, in particular, by illustrating how their metrics are affected by channel issues. It is also provided a Survey on possible orbits and constellations with a focus on devices and specific patterns of interaction between them. The paper overviews SDN, which is considered crucial for controlling QNs.

Moreover, some feasibility studies on quantum NTNs using satellites and aerial platforms are described. The first scenarios that are examined concern the measurement of some parameters on E2E paths involving single satellites. In addition, the implementation of satellite backbones through the use of SDN are also explored. Specifically, in the considered scenario, the SDN Controller is located on the ground or within the constellation itself in multi-Controller configurations. Finally, studies on HAPs and drones are also examined. The most significant parameters are measured, and the use of an SDN Controller integrated in one of the drones that control the path is highlighted.

The last part explains some experiments related to specific applications that could be extended on a global scale through quantum NTNs. These networks could provide most of the promising mechanisms of the future QI, such as secure communications and a distributed computational power that greatly exceeds the capacity of the classical Internet. Many experiments have been conducted in the field of security also in the NTN domain with the purpose of exchanging quantum keys considering various QKD protocols. Although some studies talk about the possibility of building a QCloud trying to provide some guidelines, there is still no definitive architectural scheme regarding this technology. However, in this survey, the possibility of connecting QCs over great distances via a satellite backbone managed through SDN in order to enable distributed computing among QDCs over great distances is discussed. Furthermore, other fundamental applications that a QSN could enable are network clock synchronization, and consequently a satellite positioning system much more precise than the current one. In addition to being used as a real-time clock for the international time scale, the proposed QN can be also used as a large-scale quantum sensor in order to probe and better understand many fundamental laws of physics, like general relativity law, the properties of space–time, and quantum physics.

Despite the fact that innumerable advances have been made since the launch of the first quantum satellites that have performed QKD experiments between two remote GSs, the creation of the QI satellite technology still requires additional research efforts. Therefore, multidisciplinary investigations will be required in order to pave the way for the realization of a QSB capable of enabling all the QI functionalities. It is necessary to adequately design the CP, developing both centralized path selection algorithms and appropriate communication protocols based on SDN in order to meet the requirements of the offered services. These systems are supposed to provide for the integration between terrestrial and satellite networks in order to ensure their interoperability. To this purpose, it is also necessary to further investigate the atmospheric link budget in different conditions and consider the integration of platforms such as LAP, HAP, or VLO satellites, which can allow mitigating the distance problem obtaining the best performance. Therefore, the use of quantum communication satellites for deep space communications and their accurate deployment can offer remarkable benefits and performance towards the realization of a global QI.

CRediT authorship contribution statement

Francesco Chiti: Methodology, Conceptualization. **Roberto Picchi:** Methodology, Conceptualization. **Laura Pierucci:** Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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